



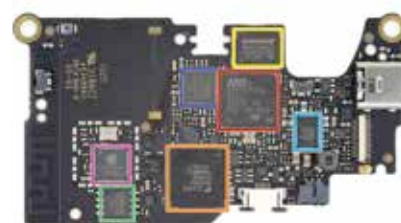
SPECS AND FEATURES

The Playdate is yellow, quirky and fun! It has a 2.7-inch display with 400x240 resolution and 173 PPI. You get a 180 MHz ARM Cortex-M7F CPU paired with 16 MB of onboard memory and 4 GB flash storage. The console also has 2.4 GHz Wi-Fi support, Bluetooth v4.2, USB-C connectivity, and a headphone jack. Oh, and a mechanical crank, because why not? The crank serves as a control for some games BTW. No, it doesn't charge the device.



RELATIVELY SIMPLE BATTERY REMOVAL PROCESS

The battery of the PlayDate is relatively easy to remove out of the body. It is glued down but the glue isn't very strong and comes off pretty easily. The battery clocks in at 2.74 Wh (740 mAh @ 3.7V). This equates to about 25 per cent of the battery capacity of the iPhone 12, as pointed out by iFixit, so that's pretty reassuring for a small handheld console.



CHIPS POWERING THE PLAYDATE

As mentioned above, you have an ARM Cortex-M7 based SoC, 4 GB of Kioxia (formerly Toshiba) NAND flash memory, 128 MB of Winbond low-power SRAM, 32 MB of Winbond NOR flash memory, a power management IC, Cirrus Logic audio codec w/ headphone/speaker amplifiers, and Espressif Systems Wi-Fi and Bluetooth SoC. There are numerous other chips present on the board as well, you can check those out on iFixit's website if you wish.



OPENING UP THE PLAYDATE

On to opening up this Game Boy-esque console! The screws holding the back panel come off pretty easily using a flathead screwdriver. Next, there are a few plastic clips that guard the innards. You can open these up with a triangular opening tool. The first thing you see when you open up the Playdate is tidy-looking insides and an easily accessible battery. There's a warranty sticker on the battery that tells you breaking it will void your warranty though.



DIGGING INTO MORE COMPONENTS

Most of the components can be removed from inside the body for repairs. Everything topples out of the body once you use a driver and spudger, and remove some ribbon cables. You can also see how the crank inside the console works. There are no gears at all. When you remove it, you can see a cylindrical magnet embedded in the shaft. As it cranks, the rotating magnetic field hits the Infineon TLV493D-A1B6 sensor which activates a particular control in your game.



SLIGHTLY TROUBLESOME DISPLAY

The 2.7-inch display is firmly glued to the front half of the console's plastic enclosure. So, if you need to replace the display, you'll probably need to replace the whole front plastic face since it is hard to separate the screen without destroying it. So, the Playdate is relatively repairable since the battery and many components are easily accessible. However, the display is glued to the plastic frame and the USB-C port is soldered onto the main board. *